Lesson 04 Demo 03

Implementing Route Parameters in a React Application

Objective: To demonstrate the implementation of route parameters in a React application using Vite, React Router, and components for enhanced navigation and dynamic content

Tools required: Visual Studio Code and Node.js

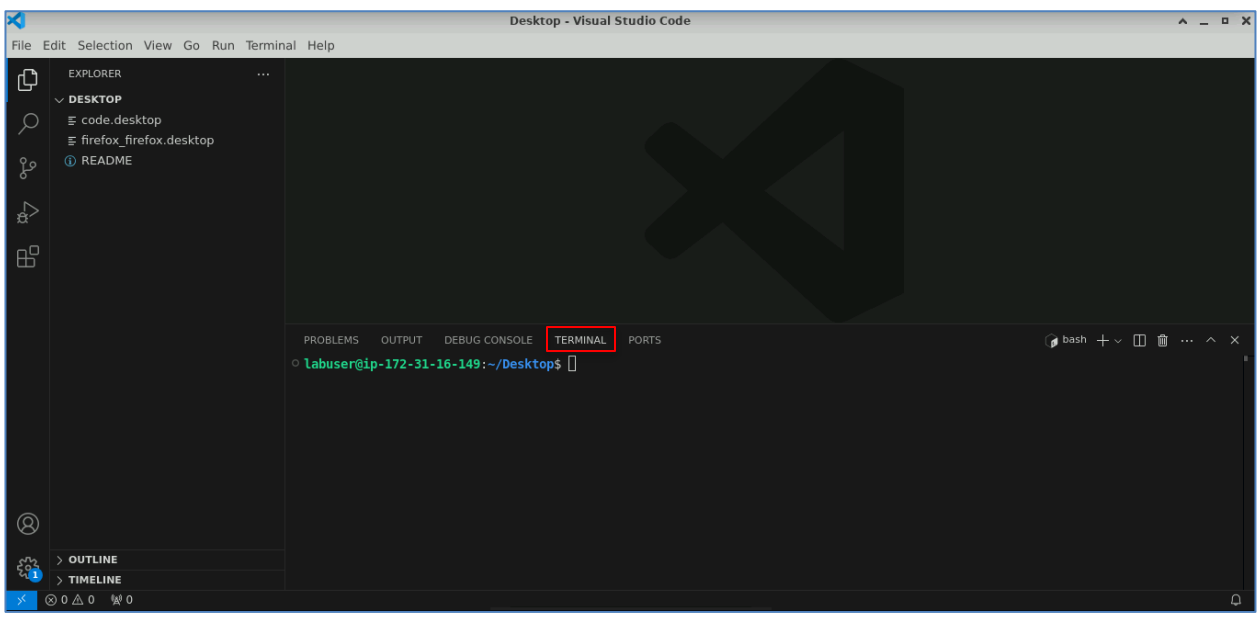
Prerequisites: None

Steps to be followed:

1. Set up a new React project using Vite
2. Create React components
3. Import and use the components
4. Run and verify the application

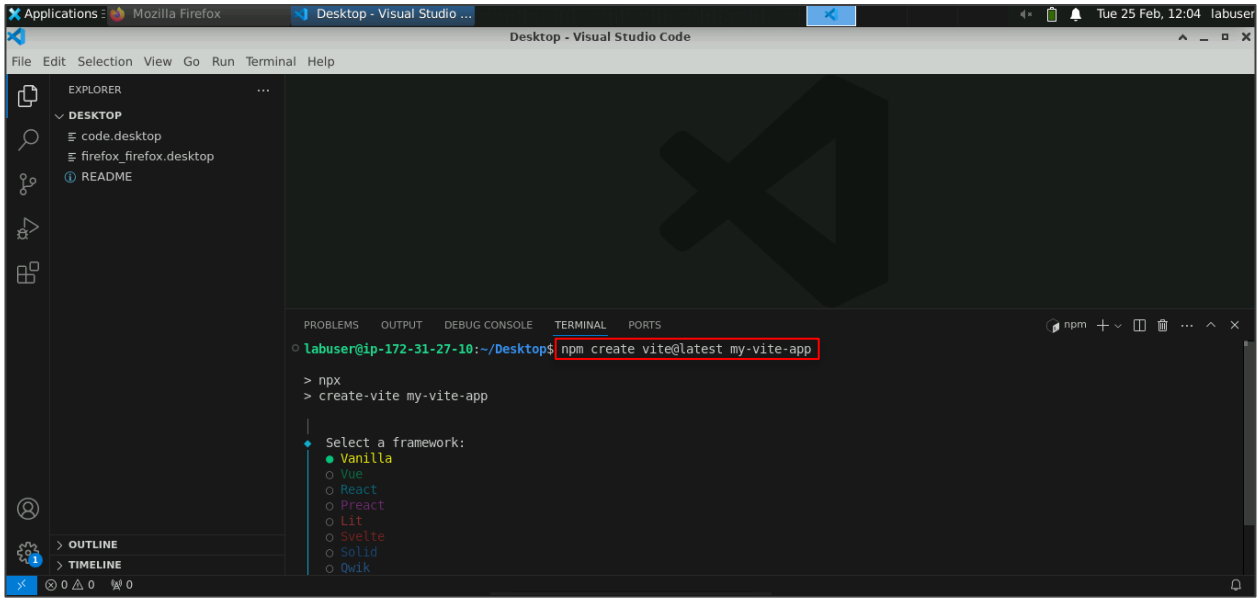
Step 1: Set up a new React project using Vite

1.1 Navigate to VS Code and open the terminal

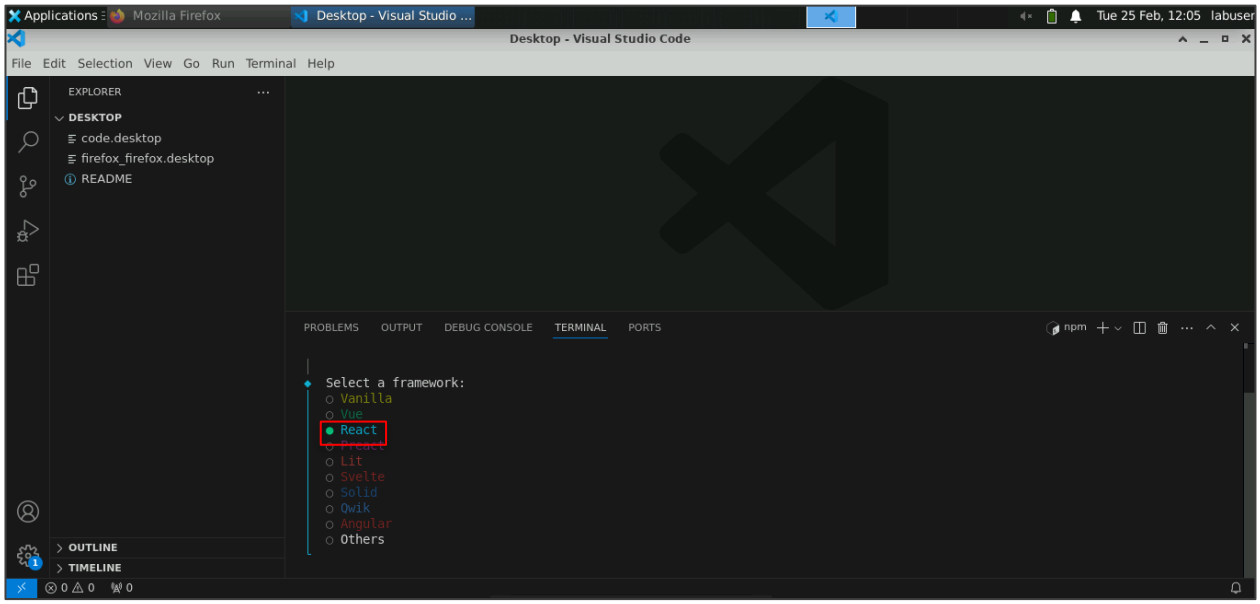


1.2 Run the following command to create a new React project using Vite:

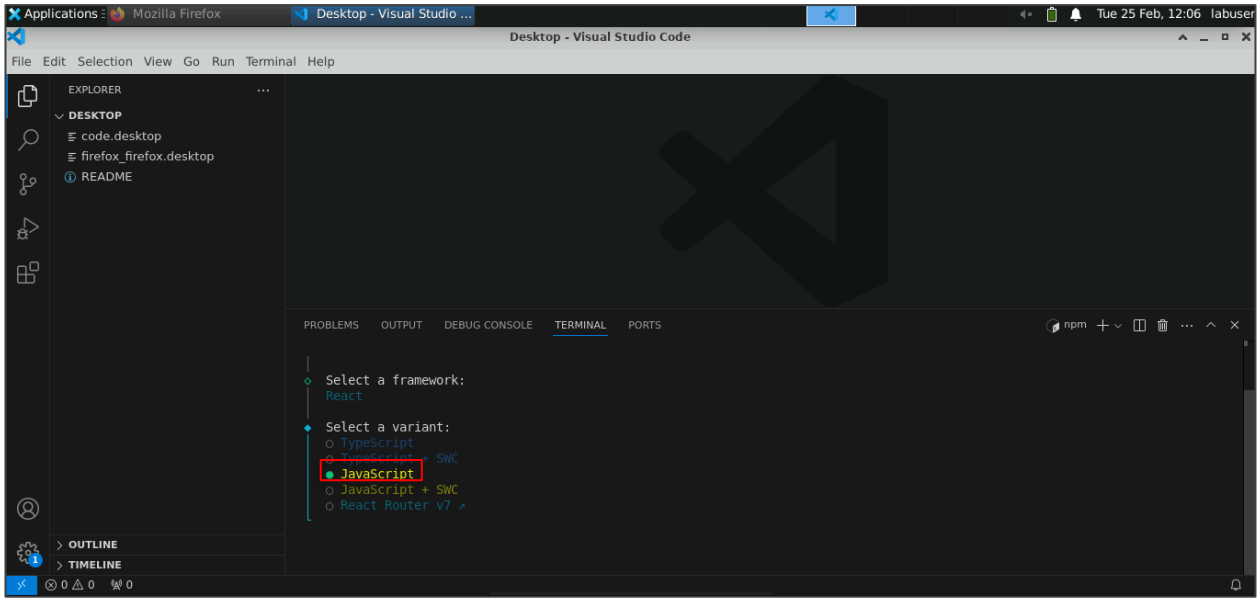
npm create vite@latest my-vite-app



1.3 Select **React** and press enter

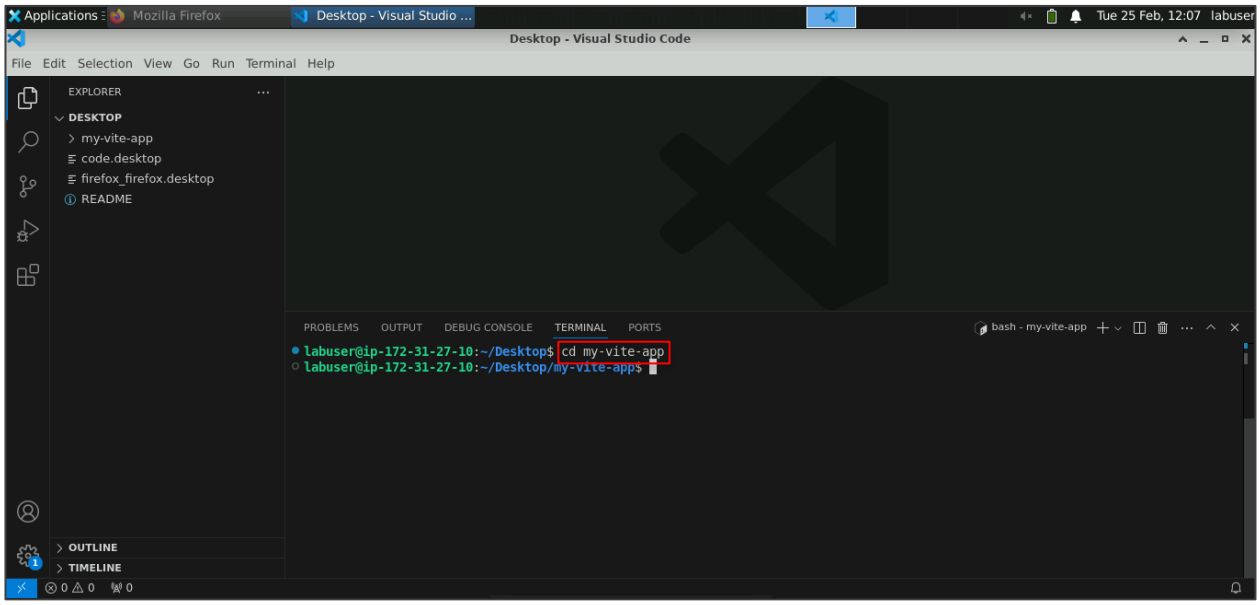


1.4 Select **JavaScript** and press enter



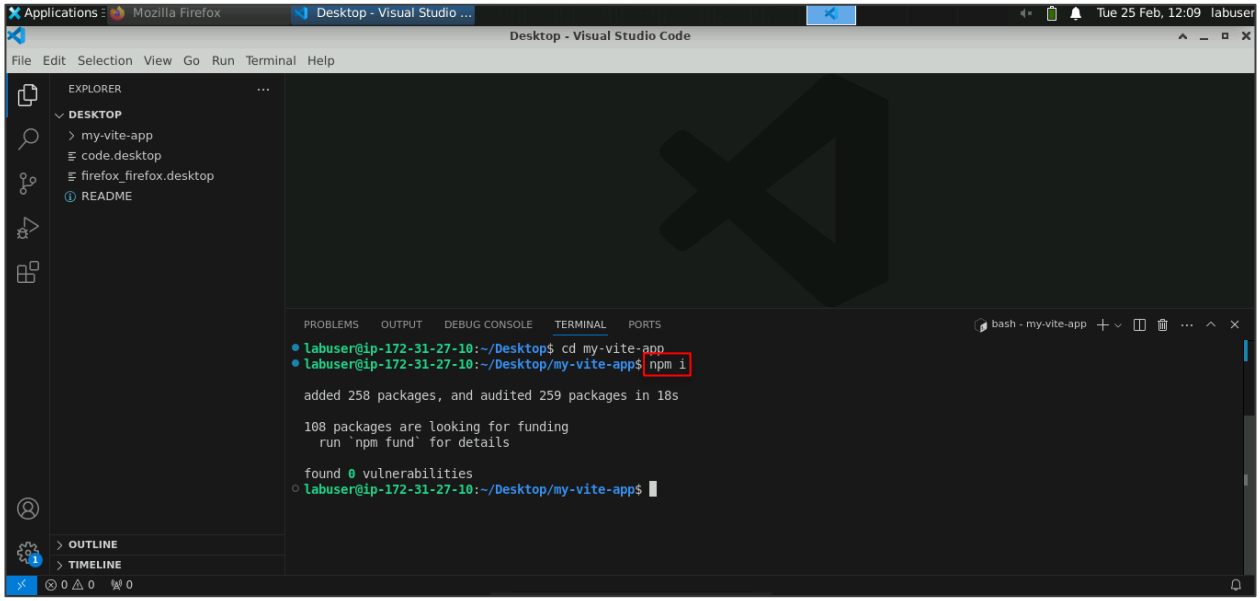
1.5 Run the following command to navigate to the project folder:

cd my-vite-app



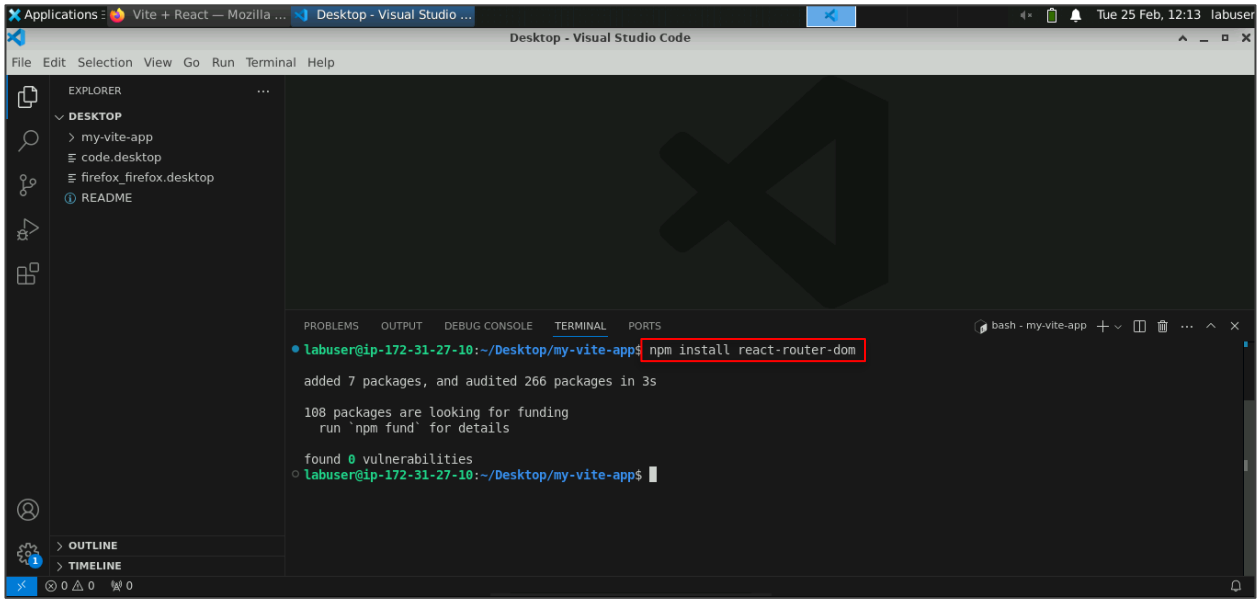
1.6 Run the following command to install dependencies:

npm i



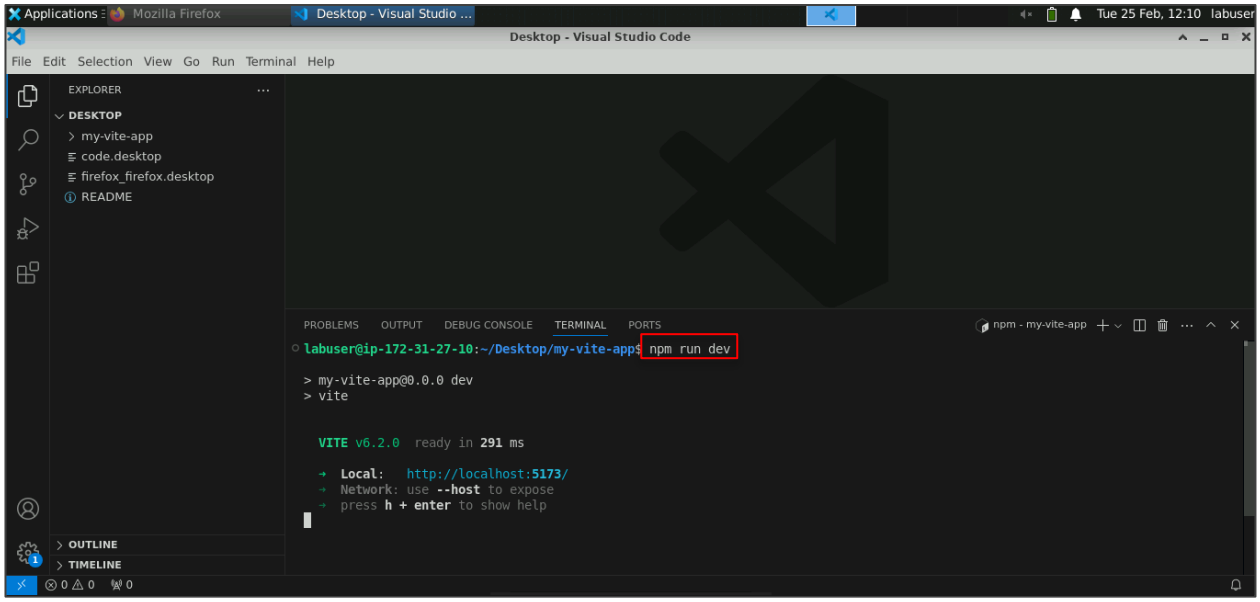
1.7 Run the following command to install the React router:

npm install react-router-dom



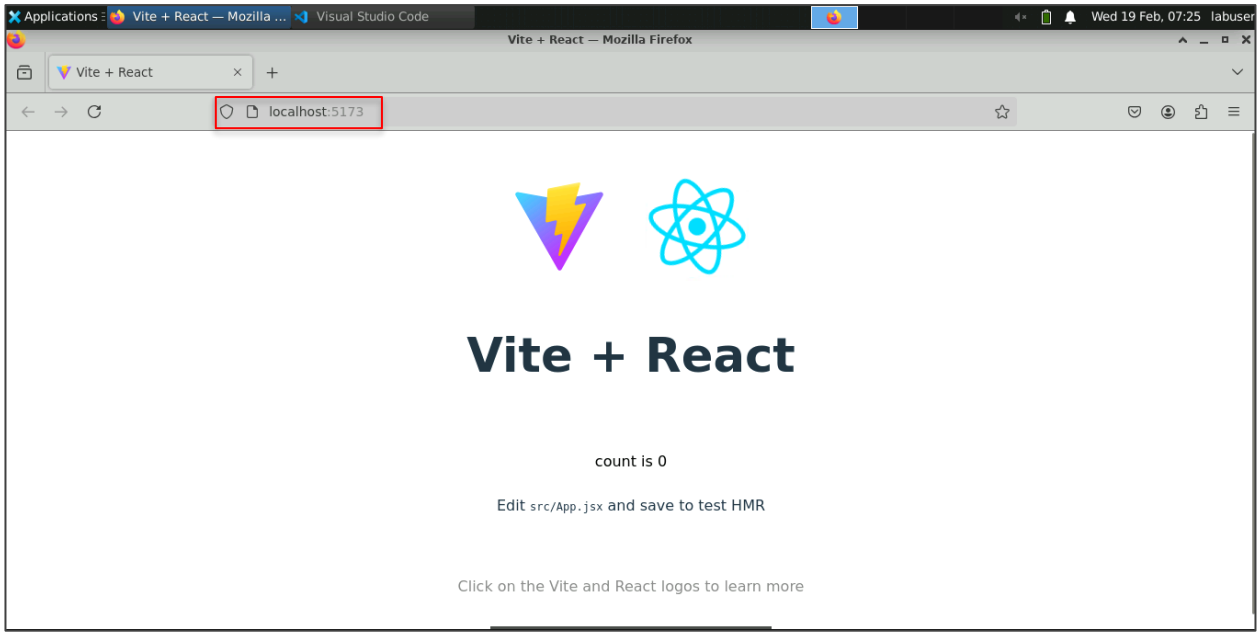
1.8 Run the following command to start the development server:

npm run dev



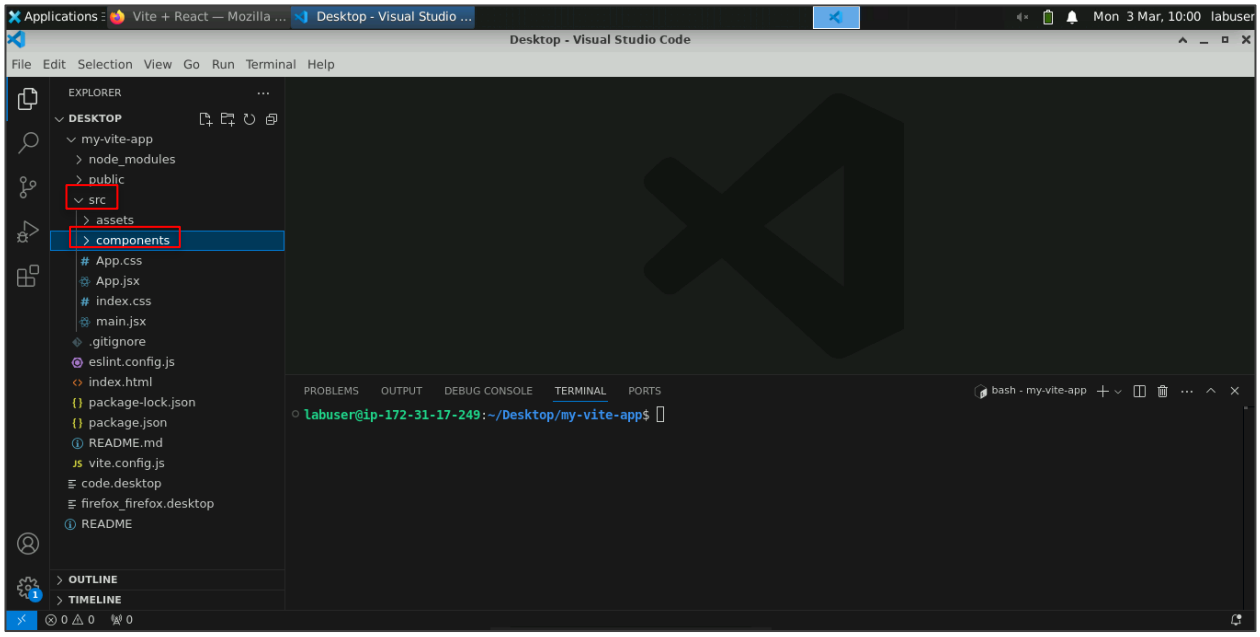
1.9 Navigate to the web browser and paste the link below:

http://localhost:5173

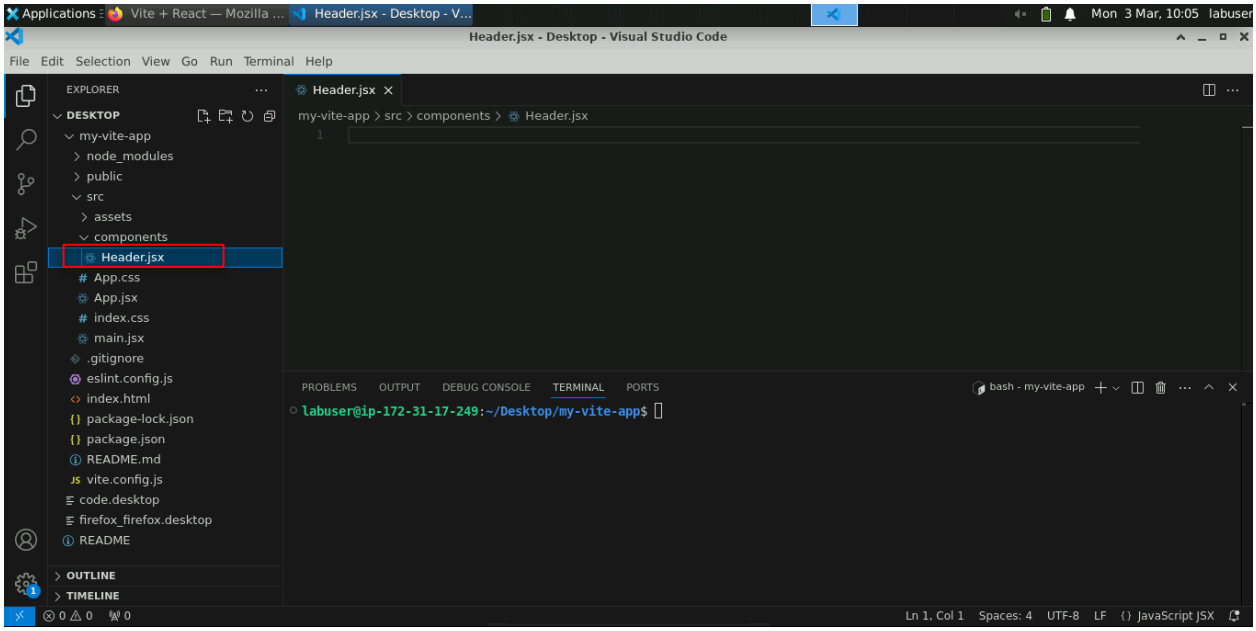


Step 2: Create React components

2.1 Inside the **src** folder, create a new folder named **components**

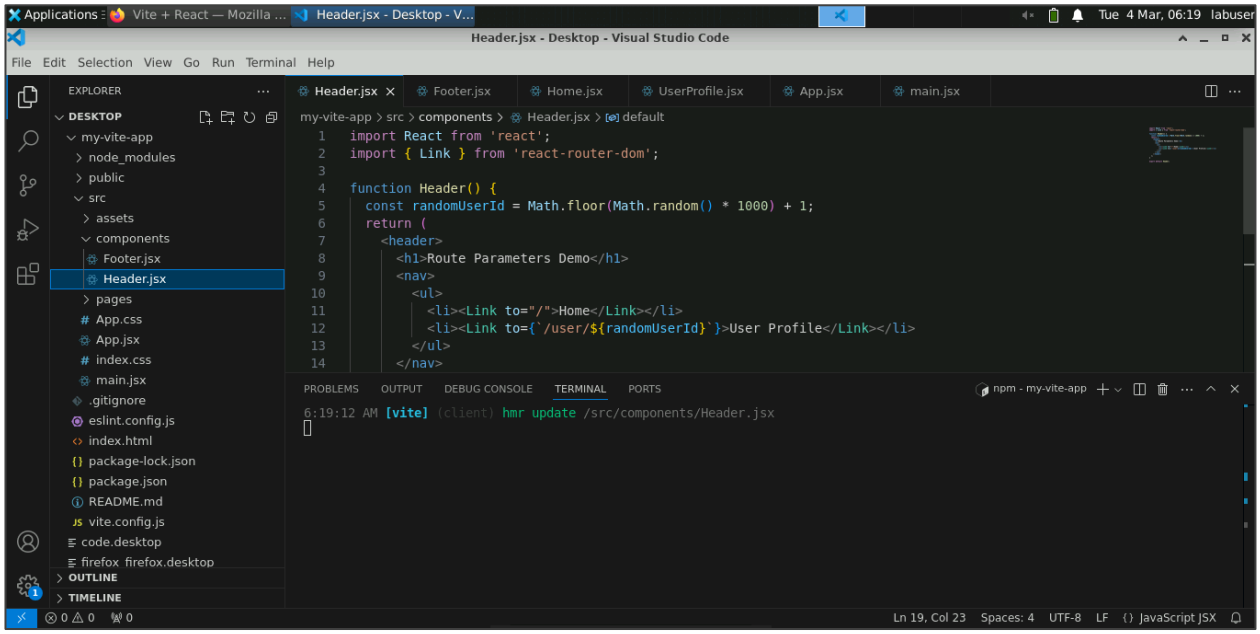


2.2 Inside the **components** folder, create a new file named **Header.jsx**

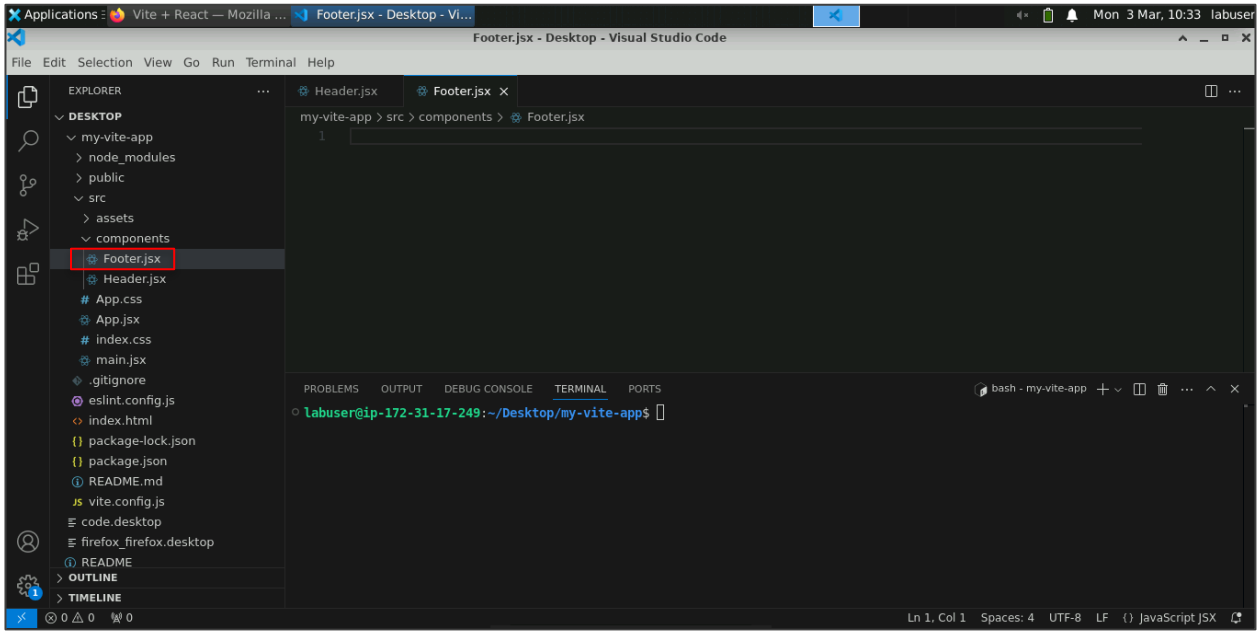


2.3 Add the following code to **Header.jsx:**

```
import React from 'react';
import { Link } from 'react-router-dom';

function Header() {
  const randomUserId = Math.floor(Math.random() * 1000) + 1;
  return (
    <header>
      <h1>Route Parameters Demo</h1>
      <nav>
        <ul>
          <li><Link to="/">Home</Link></li>
          <li><Link to={`/user/${randomUserId}`}>User Profile</Link></li>
        </ul>
      </nav>
    </header>
  );
}

export default Header;
```

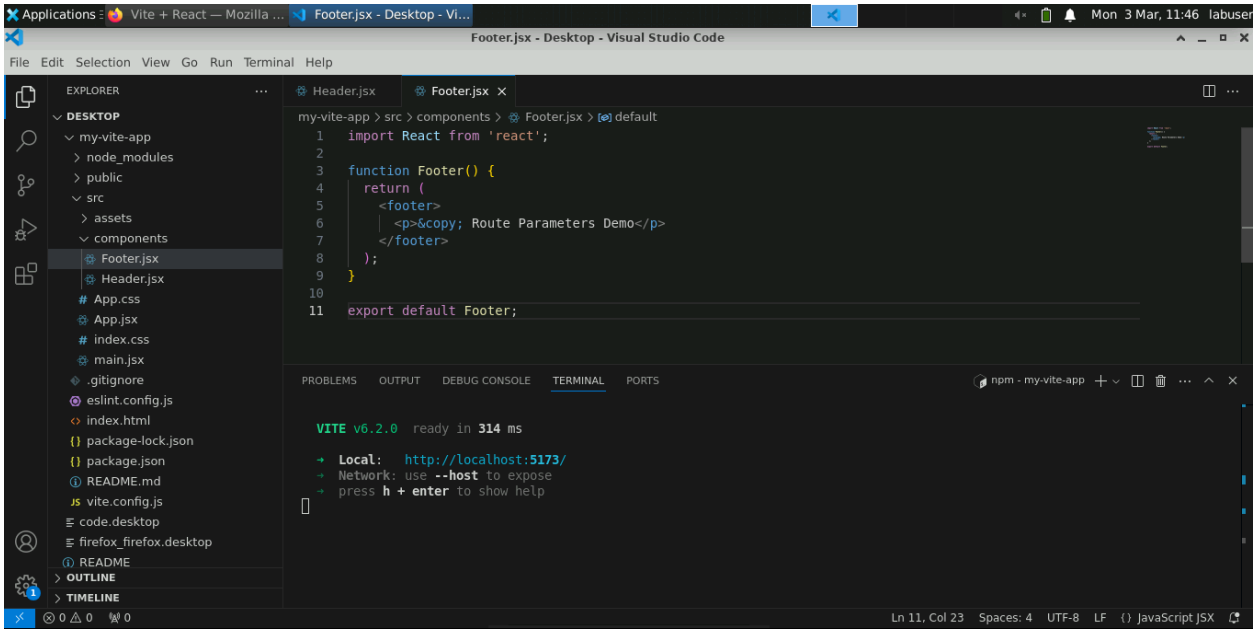


2.4 Inside the **components** folder, create a new file named **Footer.jsx**

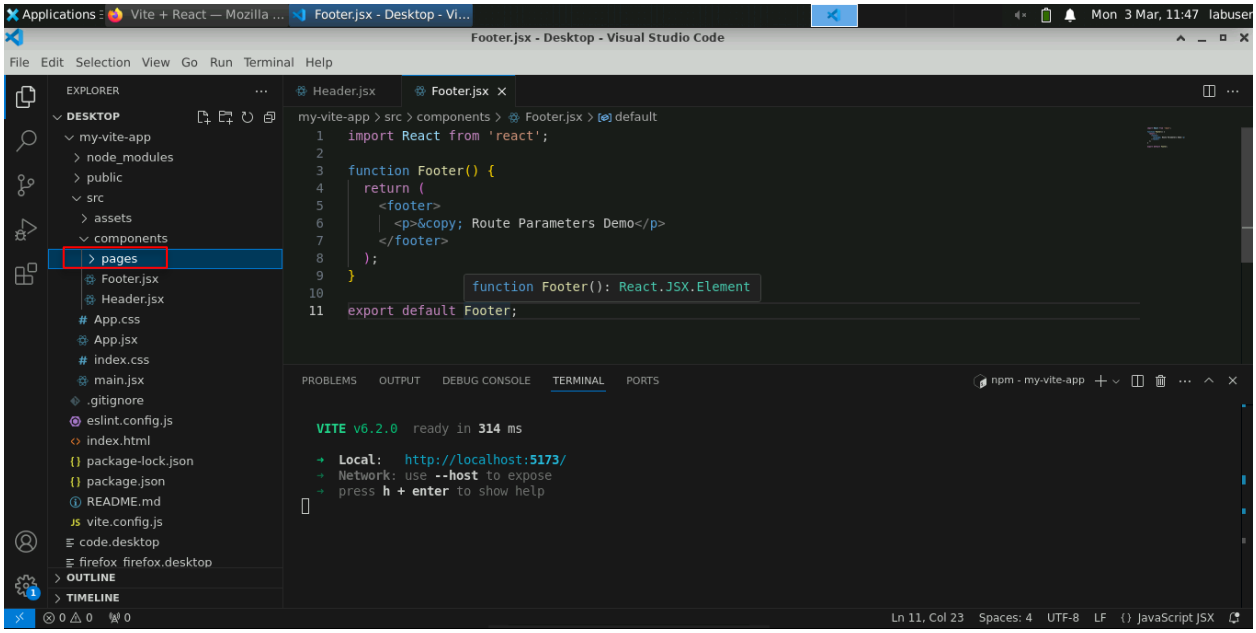


2.5 Add the following code to **Footer.jsx:**

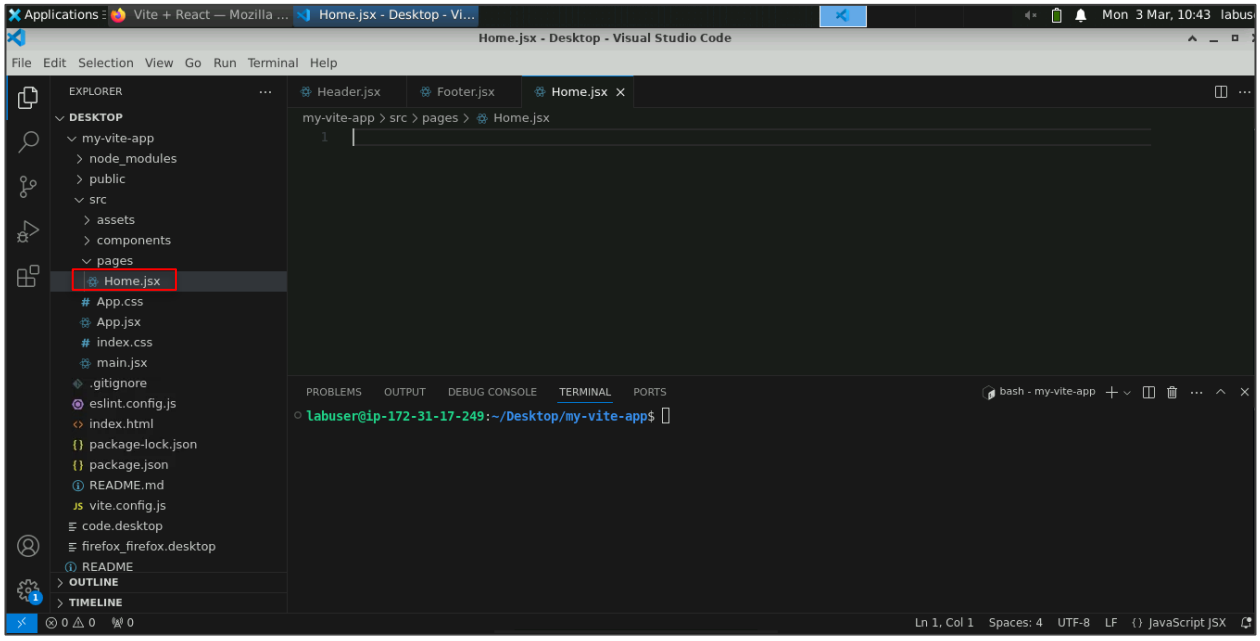
```
import React from 'react';

function Footer() {
  return (
    <footer>
      <p>&copy; Route Parameters Demo</p>
    </footer>
  );
}

export default Footer;
```



2.6 Inside the **src** folder, create a new folder named **pages**

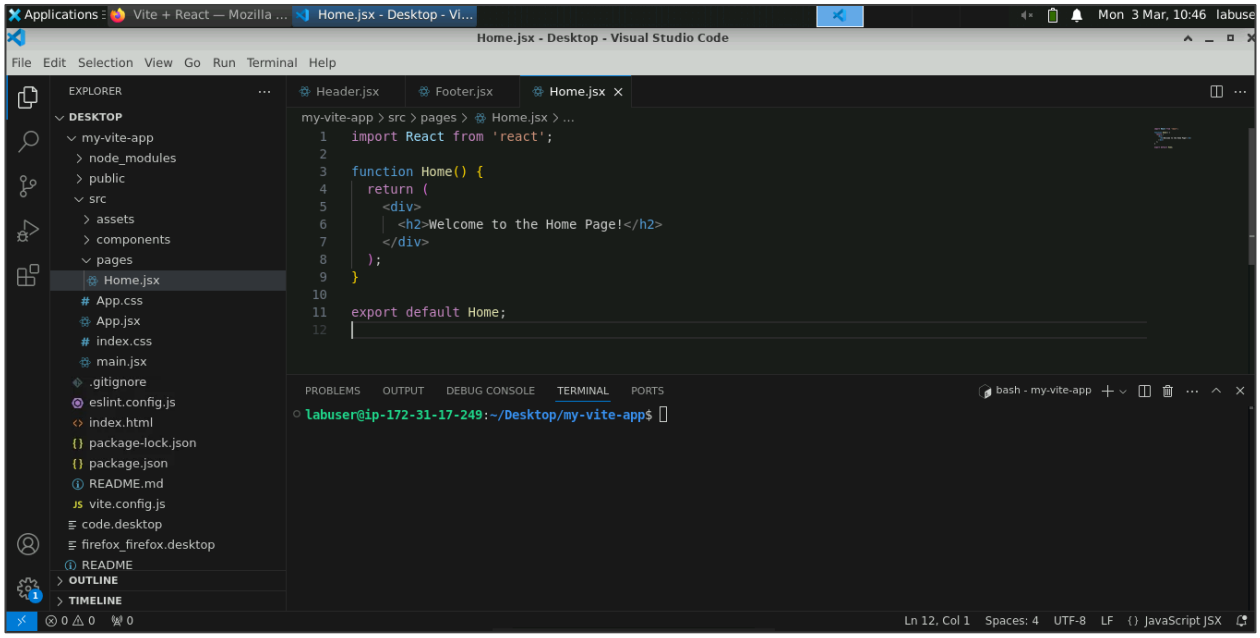


2.7 Inside the **pages** folder, create a new file named **Home.jsx**

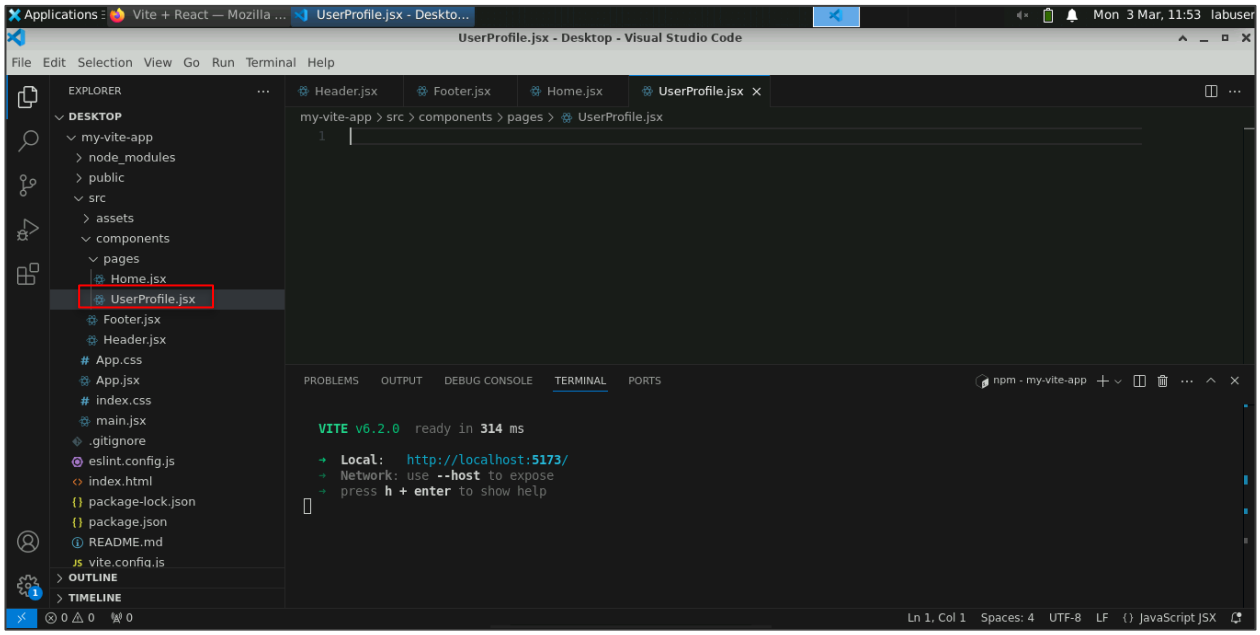


2.8 Add the following code to **Home.jsx:**

```
import React from 'react';

function Home() {
  return (
    <div>
      <h2>Welcome to the Home Page!</h2>
    </div>
  );
}

export default Home;
```

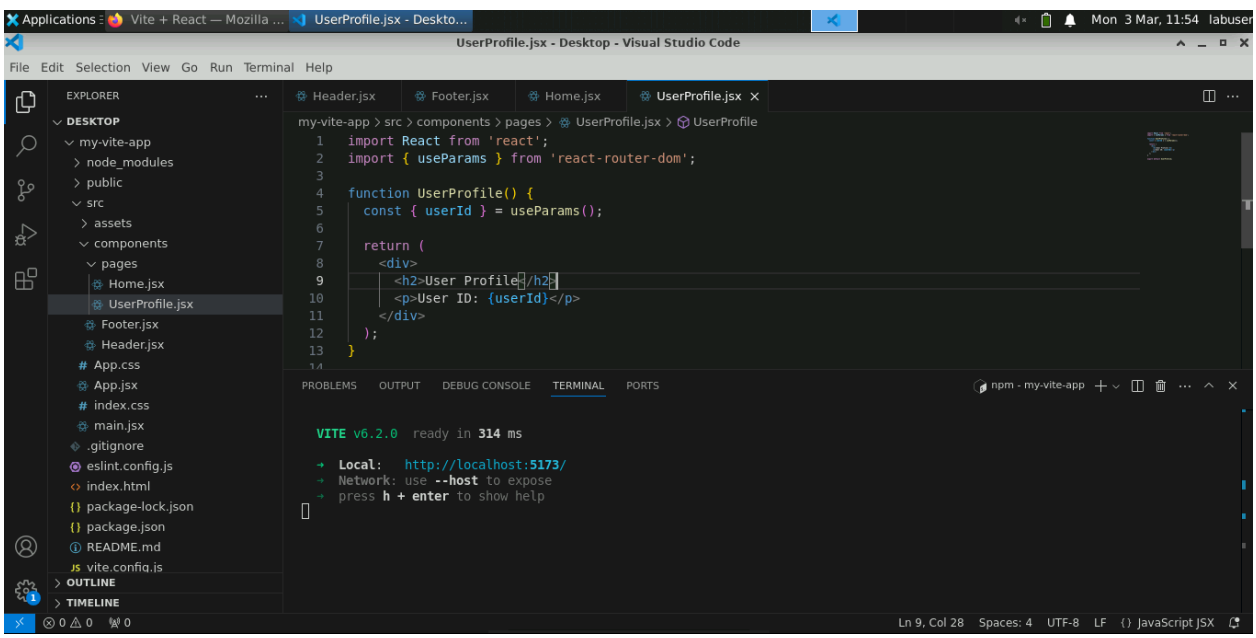


2.9 Inside the **pages** folder, create a new file named **UserProfile.jsx**



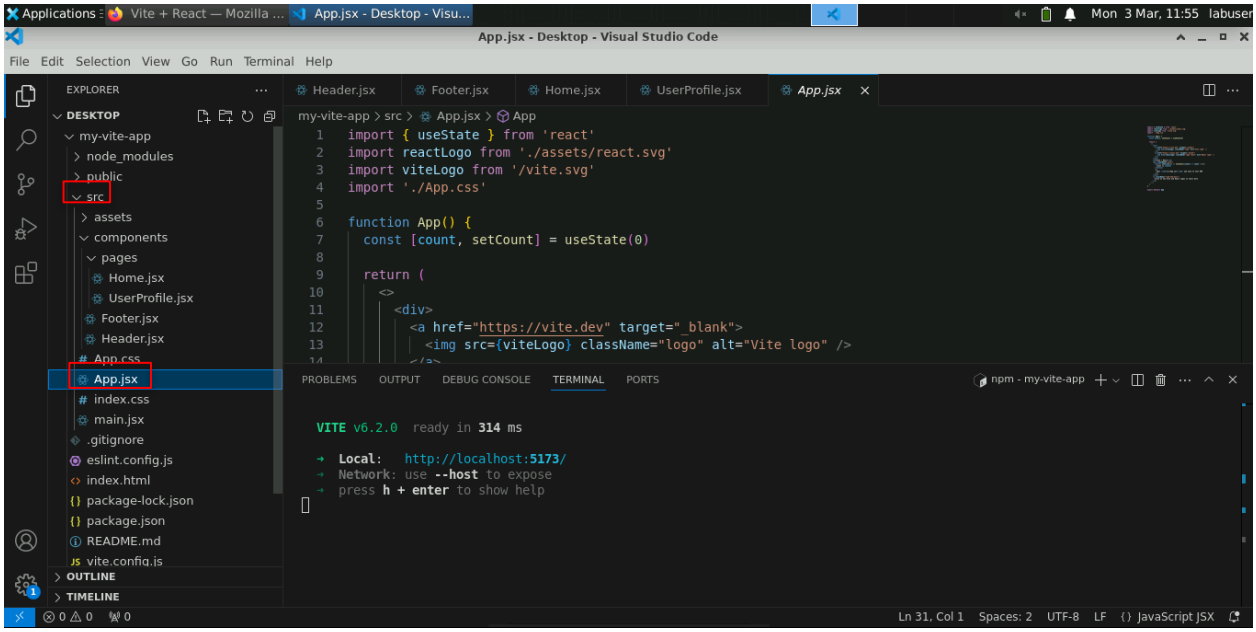
2.10 Add the following code to **UserProfile.jsx:**

```
import React from 'react';
import { useParams } from 'react-router-dom';

function UserProfile() {
  const { userId } = useParams();

  return (
    <div>
      <h2>User Profile</h2>
      <p>User ID: {userId}</p>
    </div>
  );
}

export default UserProfile;
```



Step 3: Import and use the components

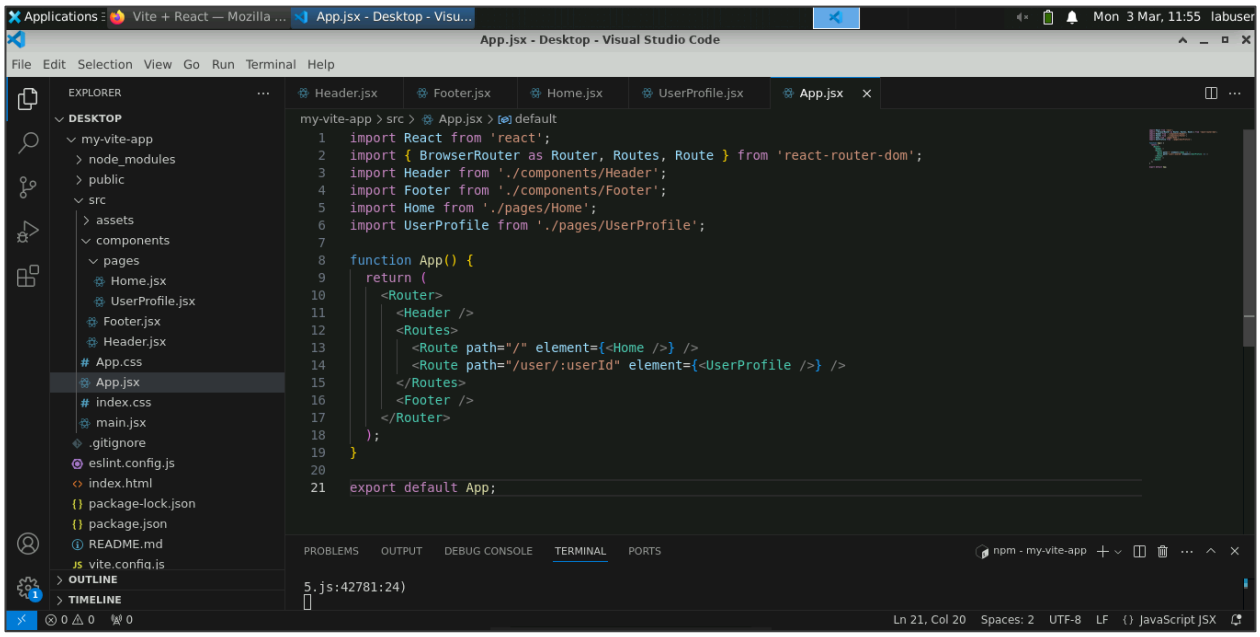
3.1 Open the **App.jsx** file inside the **src** folder



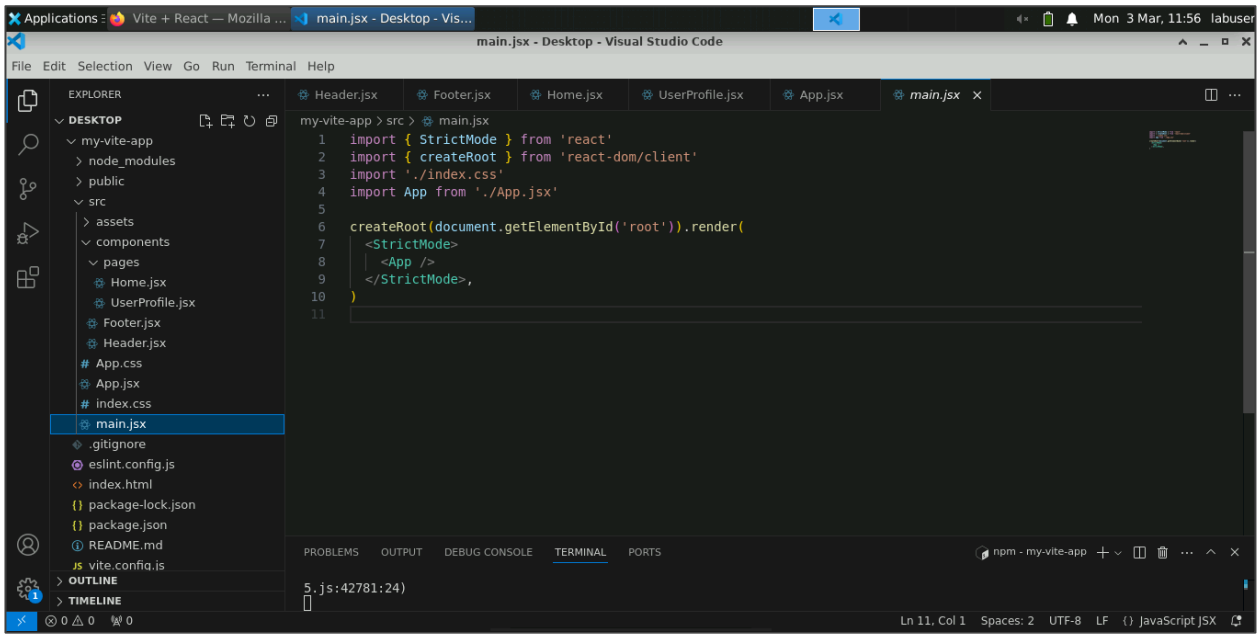
3.2 Paste the code below to include the new components in **App.jsx**:

```
import React from 'react';
import { BrowserRouter as Router, Routes, Route } from 'react-router-dom';
import Header from './components/Header';
import Footer from './components/Footer';
import Home from './pages/Home';
import UserProfile from './pages/UserProfile';

function App() {
  return (
    <Router>
      <Header />
      <Routes>
        <Route path="/" element={<Home />} />
        <Route path="/user/:userId" element={<UserProfile />} />
      </Routes>
      <Footer />
    </Router>
```

```
  );
}

export default App;
```

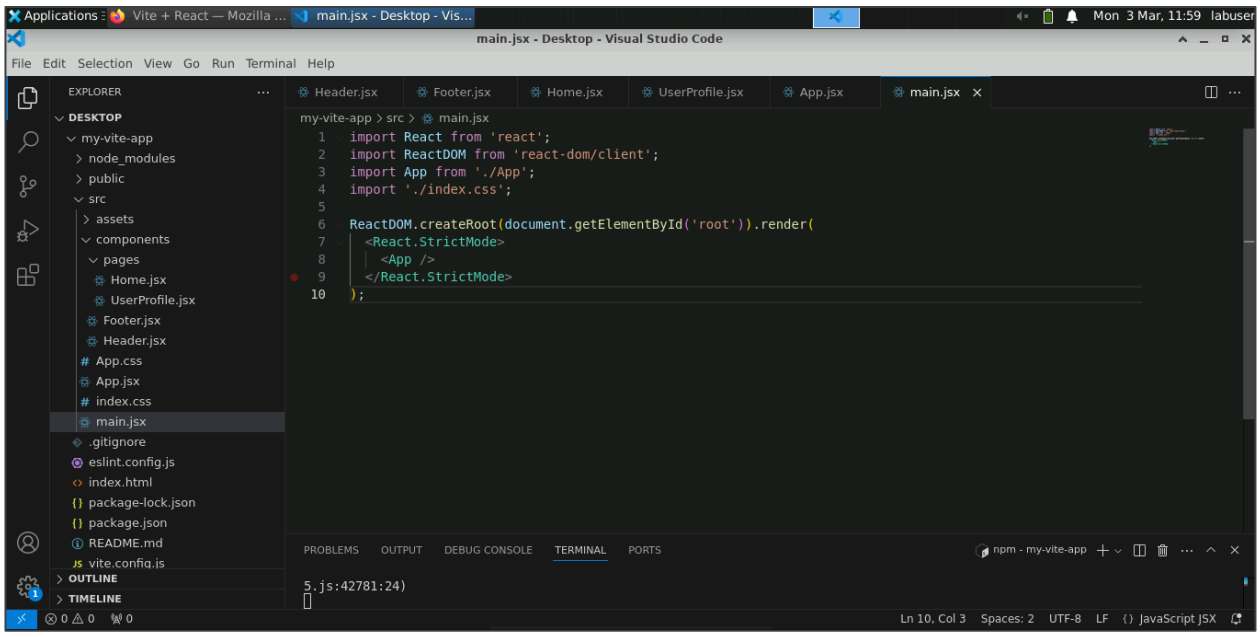


3.3 Open the **main.jsx** file inside the **src** folder



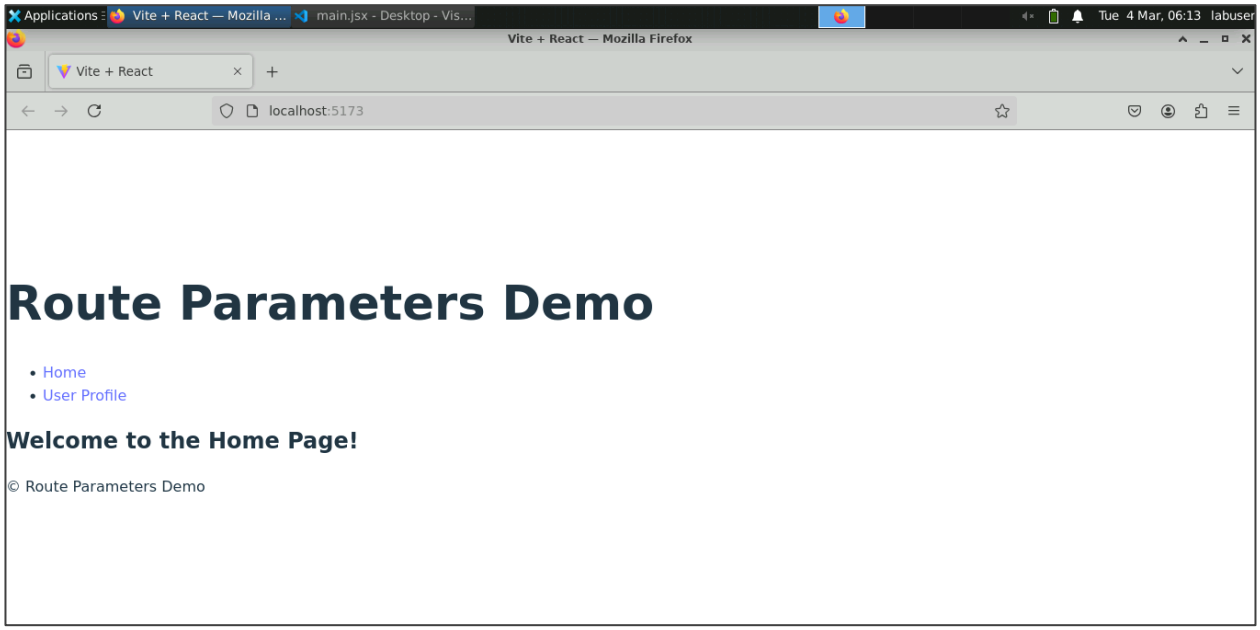
3.4 Paste the code below to include the new components in **main.jsx**:

```
import React from 'react';
import ReactDOM from 'react-dom/client';
import App from './App';
import './index.css';

ReactDOM.createRoot(document.getElementById('root')).render(
  <React.StrictMode>
    <App />
  </React.StrictMode>
);
```

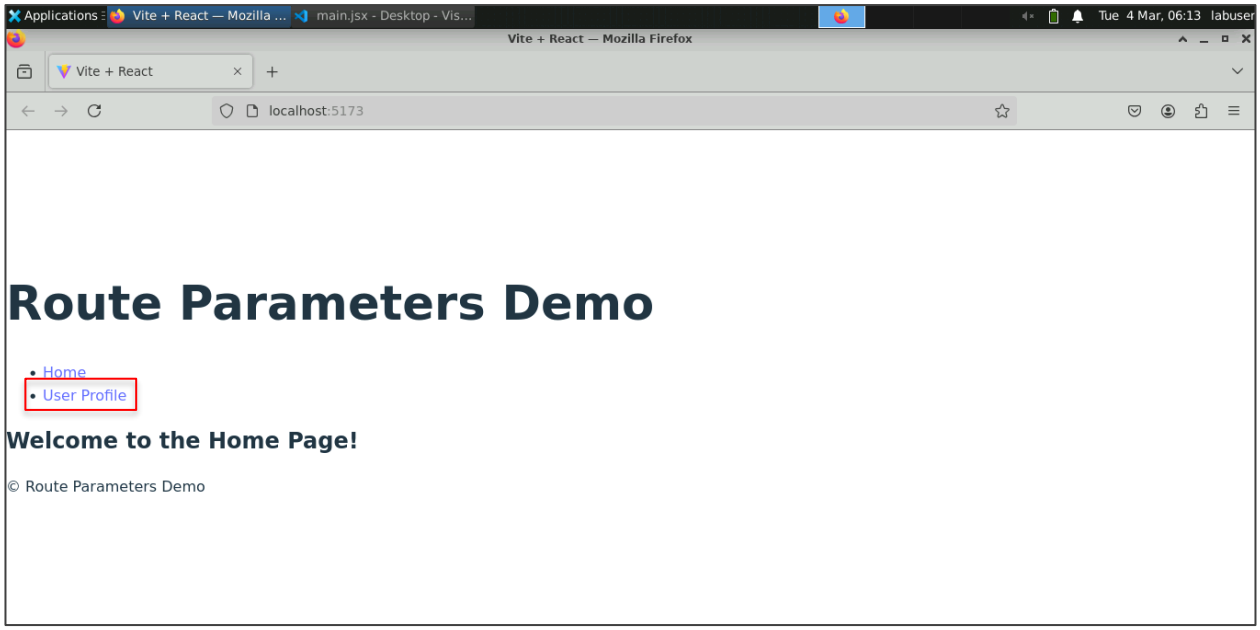


Step 4: Run and verify the application

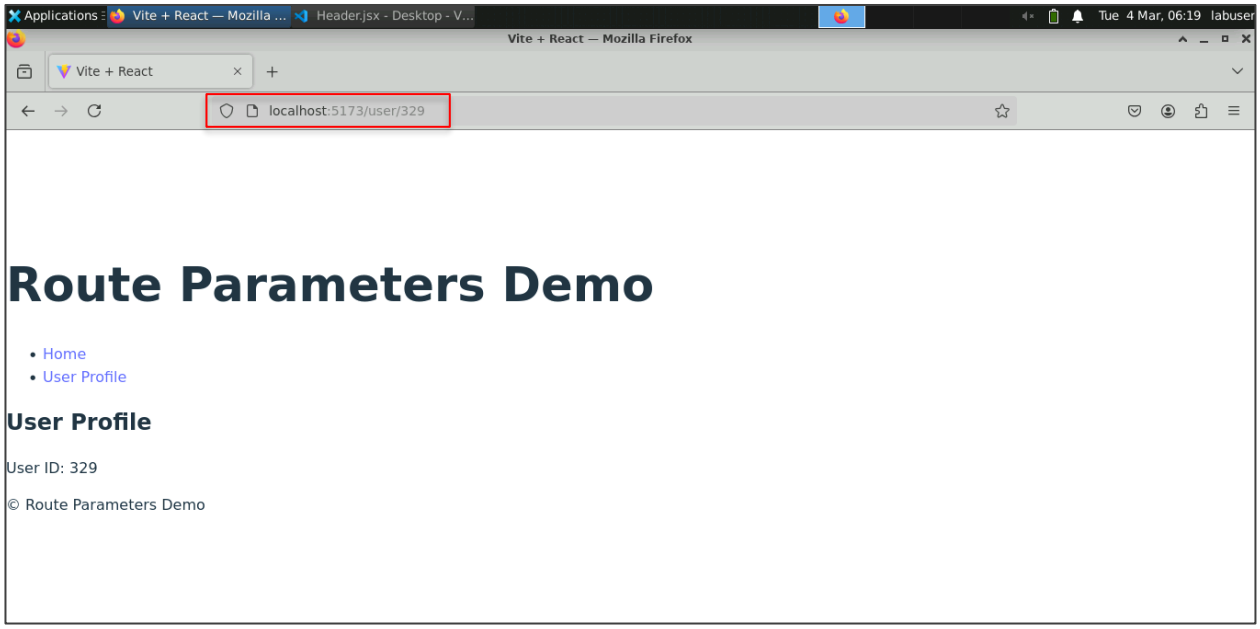
4.1 Navigate to the browser and refresh the webpage



4.2 Click **User Profile** to navigate to **/user/userId** route



Note: The User ID in the route will be randomly generated and change with each refresh.



By following these steps, you have successfully set up a React application with Vite, implemented route parameters, and improved navigation using React Router and components.